Economics 880 - PS4

Aaron Banks & Arthur Rodrigues

Question 1

The moment vector $m_3(x_t)$ contains the quantities used to construct the mean, variance, and first-order autocovariance of the time series $x \equiv \{x_t\}$. We have

$$m_3(x_t) = \begin{bmatrix} x_t \\ (x_t - \bar{x})^2 \\ (x_t - \bar{x})(x_{t-1} - \bar{x}) \end{bmatrix}.$$

We will consider the moments of the stationary distribution of the series since we are asked to compute the asymptotic moments. We have

$$\begin{aligned} \mathbb{E}[x_t] &= \rho_0 \mathbb{E}[x_{t-1}] + \mathbb{E}[\varepsilon_t] \\ &= \rho_0 \mathbb{E}[x_{t-1}]. \end{aligned}$$

Under stationarity, $\mathbb{E}[x_t] = \mu$ for all t, hence

$$\mu = \rho_0 \mu \qquad \Rightarrow \qquad \mu(1 - \rho_0) = 0 \qquad \Rightarrow \qquad \mu = 0,$$

since $\rho_0 = 0.5$.

For the asymptotic variance, we have

$$\begin{aligned} \mathbb{E}[x_t^2] &= \rho_0^2 \mathbb{E}[x_{t-1}^2] + 2\rho_0 \mathbb{E}[x_{t-1}\varepsilon_t] + \mathbb{E}[\varepsilon_t^2] \\ &= \rho_0^2 \mathbb{E}[x_{t-1}^2] + \sigma_0^2, \end{aligned}$$

since $\mathbb{E}[x_{t-1}\varepsilon_t] = 0$. Under stationarity, $\mathbb{E}[x_{t-1}^2] = \mathbb{E}[x_t^2] = \nu$, so

$$\nu = \rho_0^2 \nu + \sigma_0^2 \qquad \Rightarrow \qquad \nu = \frac{\sigma_0^2}{1 - \rho_0^2}.$$

For the first-order autocovariance, we have

$$\begin{aligned} \mathbb{E}[x_t x_{t-1}] &= \mathbb{E}[(\rho_0 x_{t-1} + \varepsilon_t) x_{t-1}] \\ &= \rho_0 \mathbb{E}[x_{t-1}^2] + \mathbb{E}[\varepsilon_t x_{t-1}] \\ &= \rho_0 \mathbb{E}[x_{t-1}^2] \\ &= \frac{\rho_0 \sigma_0^2}{1 - \rho_0^2}, \end{aligned}$$

where the third equality follows from the independence of ε_t and x_{t-1}, and the fourth from stationarity.

Thus, for the parameter vector $b = (\rho, \sigma^2)^\top$, the population moment vector is

$$\mu_3(b) = \begin{bmatrix} 0 \\ \dfrac{\sigma^2}{1-\rho^2} \\ \dfrac{\rho\sigma^2}{1-\rho^2} \end{bmatrix}.$$

Evaluating at the true parameters, $b_0 = (\rho_0, \sigma_0^2)^\top$, gives

$$\mu_3(b_0) = \begin{bmatrix} 0 \\ \dfrac{\sigma_0^2}{1-\rho_0^2} \\ \dfrac{\rho_0\sigma_0^2}{1-\rho_0^2} \end{bmatrix} = \begin{bmatrix} 0 \\ 4/3 \\ 2/3 \end{bmatrix}.$$

We next compute the Jacobian of the population moment conditions. Recall that

$$g(b) = \mu_3(b_0) - \mu_3(b),$$

so

$$\nabla_b g(b) = -\frac{\partial \mu_3(b)}{\partial b^\top}.$$

Differentiating with respect to ρ and σ^2, we obtain

$$\nabla_b g(b) = -\begin{bmatrix} 0 & 0 \\ \dfrac{2\rho\sigma^2}{(1-\rho^2)^2} & \dfrac{1}{1-\rho^2} \\ \dfrac{\sigma^2(1+\rho^2)}{(1-\rho^2)^2} & \dfrac{\rho}{1-\rho^2} \end{bmatrix}.$$

Evaluating at b_0, we have

$$G \equiv \nabla_b g(b_0) = -\begin{bmatrix} 0 & 0 \\ 16/9 & 4/3 \\ 20/9 & 2/3 \end{bmatrix}.$$

The Jacobian has full column rank since

$$\det \begin{bmatrix} 16/9 & 4/3 \\ 20/9 & 2/3 \end{bmatrix} = -\frac{16}{9} \neq 0.$$

Thus,

$$\text{rank}(G) = 2,$$

and the two structural parameters are locally identified.

We next compute the AGS sensitivity matrix, defined as

$$\Lambda = -(G^T W G)^{-1} G^T W.$$

Using $W = I_3$, we have

$$\Lambda = -(G^T G)^{-1} G^T.$$

Substituting the Jacobian and simplifying gives

$$\Lambda = \begin{bmatrix} 0 & -3/8 & 3/4 \\ 0 & 5/4 & -1 \end{bmatrix}.$$

The first column of Λ is zero, indicating that the sample mean does not affect either parameter estimate when $W = I_3$. This follows from the fact that the population mean is zero regardless of the structural parameters.

The variance and first-order autocovariance are informative for both parameters. In particular, the first row of Λ indicates that an increase in the empirical variance decreases the estimate of ρ, holding the autocovariance fixed, while an increase in the empirical autocovariance increases the estimate of ρ, holding the variance fixed.

Similarly, the second row indicates that an increase in the empirical variance increases the estimate of σ^2, holding the autocovariance fixed, while an increase in the empirical autocovariance decreases the estimate of σ^2, holding the variance fixed.

Thus, the mean provides no identifying information, while the variance and autocovariance jointly identify the two structural parameters.